

Engine Will Not Crank From the Engine Room or Remote Panel

Symptoms

- Engine will **not** crank when the start button is pushed at the engine room panel.
- Engine will **not** crank when the start button is pushed at the remote panel.

How To Use This Tree

This symptom tree can be used to troubleshoot engine start symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

- To initiate engine crank from the engine room panel, the following panel parameters **must** be met:
- The engine room panel power switch on the lamp illuminated.
 - The engine is stopped.
- To initiate engine crank from the remote panel, the following panel parameters **must** be met:
- The remote panel power lamp illuminated.
 - The local start **only** lamp is **not** illuminated
 - The engine is stopped.
- Prior to beginning the troubleshooting procedure, the following conditions **must** be active:
1. All circuit breakers in the customer interface box must be closed.
 2. The engine stop switch **must** be disengaged.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Engine Room Panel Configuration	
	STEP 1A. Check Panel Power	Is the power switch on and lamp illuminated?
	STEP 1B. Check Local Start Only Lamp	Local start only lamp illuminated?

STEPS		SPECIFICATIONS
STEP 2.	Check Remote Panel Configuration	
	STEP 2A. Check Panel Power	Local start only lamp illuminated?
	STEP 2B. Check Local Start Only Lamp	Is remote panel local start only lamp illuminated?
STEP 3.	Check Engine Room Panel Start Button	
	STEP 3A. Check Start Button Input to Customer Interface Box Logic Unit	Crank lamp illuminated?
	STEP 3B. Check Start Button Operation	Less than 10 ohms resistance?
STEP 4.	Check Remote Panel Start Button	
	STEP 4A. Check Start Button Input to Customer Interface Box Logic Unit	Crank lamp illuminated?
	STEP 4B. Check Start Button Operation	Less than 10 ohms resistance?
STEP 5.	Check Panel Wiring	
	STEP 5A. Check Engine Room Panel Wiring	Less than 10 ohms resistance?
	STEP 5A-1. Check Engine Room Power Switch Supply Wire	Less than 10 ohms resistance?
	STEP 5A-2. Check Engine Room Panel Start Supply Wire	Less than 10 ohms resistance?
STEP 6.	Check Panel System Cables	
	STEP 6A. Check Engine Room Panel Cable	Less than 10 ohms resistance?
	STEP 6B. Check Remote Panel Cable	Less than 10 ohms resistance?
	STEP 6C. Check Starter Cable Prelubrication Signals	Less than 10 ohms resistance?
	STEP 6C-1. Check Starter Cable Prelubrication System Jumper	Less than 10 ohm resistance?

STEPS		SPECIFICATIONS
	STEP 6C-2. Check Starter Cable Prelubrication System Wires	Less than 10 ohm resistance?
STEP 7.	Check Customer Interface Box Wiring	
	STEP 7A. Check the Engine Room Panel Start Supply Wire	Less than 10 ohms resistance?
	STEP 7B. Check the Remote Panel Start Supply Wire	Less than 10 ohms resistance?
	STEP 7C. Check Customer Interface Box Logic Unit Prelube Activation Signal Wire	Less than 10 ohms resistance?
	STEP 7D. Check Customer Interface Box Logic Prelube Complete Signal Wire	Less than 10 ohms resistance?
	STEP 7E. Check Customer Interface Box Logic Unit Starter Relay Switch Signal Wire	Less than 10 ohms resistance?
	STEP 7F. Check Customer Interface Box Logic Unit Starter Relay Return Wire	Less than 10 ohms resistance?
STEP 8.	Check Start Signal to Engine	
	STEP 8A. Check Starter Cable Starter Relay Switch Signals	Less than 10 ohms resistance?
STEP 9.	Check Customer Interface Box Logic Unit	
	STEP 9A. Check Customer Interface Box Logic Unit From the Engine Room Panel	24 VDC?
	STEP 9B. Check Customer Interface Box Logic Unit From the Remote Panel	24 VDC?
STEP 10.	Check Engine Room Panel	
	STEP 10A. Check Start Signal from Engine Room Panel or Remote Panel	24 VDC?

STEP 1. Check Engine Room Panel Configuration

STEP 1A. Check Panel Power

Conditions :

- Locate engine room panel.

Action	Specification/Repair	Next Step
Verify power switch is on and lamp illuminated.	Is the power switch on and lamp illuminated? YES	1B
	Is the power switch on and lamp illuminated? NO Repair : Turn on power switch and verify lamp is illuminated. If the power lamp is not on refer to the System Will Not Start After Shutdown Troubleshooting Symptom Tree.	Repair complete.

STEP 1B. Check Local Start Only Lamp

Conditions :

- Locate engine room panel
- Power switch on and lamp illuminated.

Action	Specification/Repair	Next Step
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Verify local start only lamp is illuminated.	Local start only lamp illuminated? YES	2A
	Local start only lamp illuminated? NO Repair : Push local start only button and verify lamp is illuminated. If the lamp did not illuminate refer to the Engine Room Panel Local/Remote Switch Fails to Switch to Local Troubleshooting Symptom Tree.	Repair complete.

STEP 2. Check Remote Panel Configuration

STEP 2A. Check Panel Power

Conditions : Locate remote panel.		
Action	Specification/Repair	Next Step

Verify power lamp is illuminatated.	Local start only lamp illuminated? YES	2B
	Local start only lamp illuminated? NO Repair : If the lamp did not illuminate refer to the System Will Not Start After Shutdown Troubleshooting Symptom Tree.	Repair complete.

STEP 2B. Check Local Start Only Lamp

Conditions : Remote panel power lamp illuminated.		
Action	Specification/Repair	Next Step

Verify remote panel local start only lamp is illuminated.	Is remote panel local start only lamp illuminated? YES Repair : Locate engine room panel and push local start only button and verify remote panel lamp is not illuminated. If the lamp is still illuminated refer to the Remote Panel Fails to Switch to Remote Troubleshooting Symptom Tree.	3A
	Is remote panel local start only lamp illuminated? NO Repair : Refer to the Remote Panel Fails to Switch to Local Troubleshooting Symptom Tree.	Repair complete.

STEP 3. Check Engine Room Panel Start Button

STEP 3A. Check Start Button Input to Customer Interface Box Logic Unit

Conditions :

- Locate engine room panel
- Engine room panel power lamp illuminated
- Engine room panel local start **only** lamp **not** illuminated
- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check engine start button input to customer interface box logic unit. • Press the start button. • Verify crank lamp illuminated on customer interface box logic unit.	Crank lamp illuminated? YES	7E
	Crank lamp illuminated? NO	3B

STEP 3B. Check Start Button Operation

Conditions :

- Open engine room panel
- Disconnect control panel X4 connector.

Action	Specification/Repair	Next Step
Check the start button operation. • Place one test lead on the engine room power switch supply terminal of the control panel X4 connector. • Place the other test lead on the engine room panel start supply terminal of the control panel X4 connector. • Press the start button.	Less than 10 ohms resistance? YES	4A
	Less than 10 ohms resistance? NO Repair : Replace the control panel. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 4. Check Remote Panel Start Button

STEP 4A. Check Start Button Input to Customer Interface Box Logic Unit

Conditions :

- Locate remote panel
- Remote panel power lamp illuminated
- Open customer interface box.

Action	Specification/Repair	Next Step
Check the engine start button input to customer interface box logic unit. • Press the start button. • Verify the crank lamp illuminated on the customer interface box logic unit.	Crank lamp illuminated? YES	7E
	Crank lamp illuminated? NO	4B

STEP 4B. Check Start Button Operation**Conditions :**

- Open remote panel
- Disconnect control panel X4 connector.

Action	Specification/Repair	Next Step
Check the start button operation. • Place one test lead on the remote panel start supply terminal of the control panel X4 connector. • Place the other test lead on the remote panel power switch supply terminal of the control panel X4 connector. • Press the start button.	Less than 10 ohms resistance? YES	5A
	Less than 10 ohms resistance? NO Repair : Replace the control panel. Refer to Procedure 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.

STEP 5. Check Panel Wiring

STEP 5A. Check Engine Room Panel Wiring

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
Check engine room panel wiring. • Disconnect cable C14 from the engine room panel. • Place one test lead to the engine room power switch supply pin at the C14 connector. • Place the other test lead on the engine room panel start supply pin at the C14 connector. • Press the start button.	Less than 10 ohms resistance? YES	6A
	Less than 10 ohms resistance? NO	5A-1

STEP 5A-1. Check Engine Room Power Switch Supply Wire

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
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Check engine room panel wiring. • Disconnect cable C14 from the engine room panel. • Connect on test lead on the engine room power switch supply pin on the engine room panel C14 connector. • Place the other test lead on the engine room power switch supply pin on the control panel connector.	Less than 10 ohms resistance? YES	5A-2
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 5A-2. Check Engine Room Panel Start Supply Wire

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
Check engine room panel wiring. • Disconnect cable C14 from the engine room panel. • Connect one test lead on the engine room panel start supply pin on the engine room panel C14 connector. • Place the other test lead on the engine room panel start supply pin on the control panel connector.	Less than 10 ohms resistance? YES	6A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 6. Check Panel System Cables

STEP 6A. Check Engine Room Panel Cable

Conditions :

- Disconnect cable connector C14 from the engine room panel
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step
Check the engine room panel cable. • Install a jumper between engine room power switch supply pin and the engine room panel start supply pin in connector C14. • Place one test lead in the engine room panel power switch supply pin in connector C7. • Place the other test lead in the engine room panel start supply pin in connector C7.	Less than 10 ohms resistance? YES	6B
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 6B. Check Remote Panel Cable

Conditions :

- Locate and open customer interface box
- Locate and open remote panel
- Disconnect remote panel cable from customer interface box X4 connector.

Action	Specification/Repair	Next Step
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Check the remote panel cable. • Install a jumper between the remote power switch supply terminal and the remote panel start supply terminal on the remote control panel X4 connector. • Place one test lead on the remote panel power switch supply terminal in customer interface box X4 connector. • Place the other test lead on the remote panel start supply terminal in the customer interface box X4 connector.	Less than 10 ohms resistance? YES	6C
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 6C. Check Starter Cable Prelubrication Signals

Conditions :

- Disconnect cable connector C1 from the customer interface box
- Check that prelubrication system jumper is in place.

Action	Specification/Repair	Next Step
Check the starter cable. • Place one test lead on the prelubrication activation signal pin in connector C1. • Place the other test lead on the prelubrication complete signal pin in connector C1.	Less than 10 ohms resistance? YES	7A
	Less than 10 ohms resistance? NO	6C-1

STEP 6C-1. Check Starter Cable Prelubrication System Jumper

Conditions :

- Disconnect the prelubrication system jumper from the starter cable.

Action	Specification/Repair	Next Step
Check the prelubrication system jumper. • Place one test lead on the prelubrication activation signal pin in the prelubrication system jumper. • Place the other test lead on the prelubrication complete signal pin in the prelubrication system jumper.	Less than 10 ohm resistance? YES	6C-2
	Less than 10 ohm resistance? NO Repair : Replace the jumper.	Repair complete.

STEP 6C-2. Check Starter Cable Prelubrication System Wires**Conditions :**

- Disconnect cable connector C1 from the customer interface box
- Disconnect the prelubrication system jumper from the starter cable.

Action	Specification/Repair	Next Step
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Check the starter cable. • Install a jumper between the prelubrication activation signal pin and the prelubrication complete signal pin in the prelubrication system connector. • Place one test lead on the prelubrication activation signal pin in connector C1. • Place the other test lead on the prelubrication complete signal pin in connector C1.	Less than 10 ohm resistance? YES	7A
	Less than 10 ohm resistance? NO Repair : Replace the jumper.	Repair complete.

STEP 7. Check Customer Interface Box Wiring

STEP 7A. Check the Engine Room Panel Start Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the engine room panel start supply wire. - Place one test lead on the engine room panel start supply pin in connector C7. - Place the other test lead on the engine room panel start supply terminal on the customer interface box logic unit.	Less than 10 ohms resistance? YES	7B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7B. Check the Remote Panel Start Supply Wire

Conditions : - Open the customer interface box - Disconnect remote panel cable at connector X4 from the customer interface box.		
Action	Specification/Repair	Next Step
Check the remote panel start supply wire. - Place one test lead on the remote panel start supply wire terminal in customer interface box X4 connector. - Place the other test lead on the remote panel start supply wire terminal on the customer interface box logic unit.	Less than 10 ohms resistance? YES	7C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7C. Check Customer Interface Box Logic Unit Prelube Activation Signal Wire

Conditions :

- Open the customer interface box
- Disconnect cable C1 connector from the connector interface box.

Action	Specification/Repair	Next Step
Check the prelube activation signal wire. • Place one test lead on the prelube activation signal terminal on the customer interface box logic unit. • Place the other test lead in the prelube activation signal pin in the C1 connector.	Less than 10 ohms resistance? YES	7D
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7D. Check Customer Interface Box Logic Prelube Complete Signal Wire**Conditions :**

- Open the customer interface box
- Disconnect cable C1 connector from the customer interface box.

Action	Specification/Repair	Next Step
Check the prelube complete signal wire. • Place one test lead on the prelube complete signal terminal on the customer interface box logic unit. • Place the other test lead in the prelube complete signal pin in the C1 connector.	Less than 10 ohms resistance? YES	7E
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7E. Check Customer Interface Box Logic Unit Starter Relay Switch Signal Wire

Conditions :

- Open the customer interface box
- Disconnect cable C1 connector from the customer interface box.

Action	Specification/Repair	Next Step
Check the start relay switch signal wire. • Place one test lead on the starter relay switch signal terminal on the customer interface box logic unit. • Place the other test lead in the starter relay switch signal pin in the C1 connector.	Less than 10 ohms resistance? YES	7F
	Less than 10 ohms resistance? NO Repair : Replace the customer interface box logic unit. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 7F. Check Customer Interface Box Logic Unit Starter Relay Return Wire

Conditions :

- Open the customer interface box
- Disconnect cable C1 connector from the customer interface box.

Action	Specification/Repair	Next Step
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Check the starter relay return wire. - Place one test lead on the starter relay return terminal on the customer interface box logic unit. - Place the other test lead in the starter relay return pin in the C1 connector.	Less than 10 ohms resistance? YES	8A
	Less than 10 ohms resistance? NO Repair : Replace the customer interface box logic unit. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 8. Check Start Signal to Engine

STEP 8A. Check Starter Cable Starter Relay Switch Signals

Conditions :

- Disconnect cable connector C1 from the customer interface box
- Disconnect cable ring terminals from the starter relay switch.

Action	Specification/Repair	Next Step
Check the starter cable. - Install a jumper between the starter relay switch signal pin and the starter relay switch return pin in connector C1. - Place one test lead on the starter relay switch signal ring terminal. - Place the other test lead on the starter relay switch return ring terminal.	Less than 10 ohms resistance? YES	9A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 9. Check Customer Interface Box Logic Unit

STEP 9A. Check Customer Interface Box Logic Unit From the Engine Room Panel

Conditions :

- Open the customer interface box
- Disconnect cable C1 connector from the customer interface box.

Action	Specification/Repair	Next Step
Check the engine room panel start supply wire. • Place the positive test lead on the starter relay switch signal terminal on the customer interface box logic unit. • Place the negative test lead on the starter relay switch return terminal on the customer interface box logic unit. • Press the start button at the engine room panel.	24 VDC? YES	9B
	24 VDC? NO Repair : Replace the customer interface box logic unit. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 9B. Check Customer Interface Box Logic Unit From the Remote Panel

Conditions :

- Open the customer interface box
- Basic system is configured for start from remote panel (local start only lamp is not illuminated)
- Disconnect cable C1 connector from the customer interface box.

Action	Specification/Repair	Next Step
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Check the remote panel start supply wire. - Place the positive test lead on the starter relay switch signal terminal on the customer interface box logic unit. - Place the negative test lead on the starter relay switch return terminal on the customer interface box logic unit. - Press the start button at the remote panel.	24 VDC? YES	10A
	24 VDC? NO Repair : Replace the customer interface box logic unit. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 10. Check Engine Room Panel

STEP 10A. Check Start Signal from Engine Room Panel or Remote Panel

Conditions :

- Configure panel system to start from remote or engine room panel.

Action	Specification/Repair	Next Step
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Check start signal from engine room panel or remote panel. • Place the positive test lead on the starter relay switch signal ring terminal. • Place the negative test lead on the starter relay switch return ring terminal. • Press the start button.	24 VDC? YES Repair : Refer to Service Manual, QSK19 and QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021592 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021592.html).	Repair complete.
	24 VDC? NO Repair : Replace the cable.	Repair complete.